CS 240 Programming Assignment Phase 2

Winter Semester 2025-2026

Presenter: Xiradakis Ioannis – xiradakis@csd.uoc.gr

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Phase 2 Concept

Main Idea: Extension of Library Management System

Extension of Library Management System

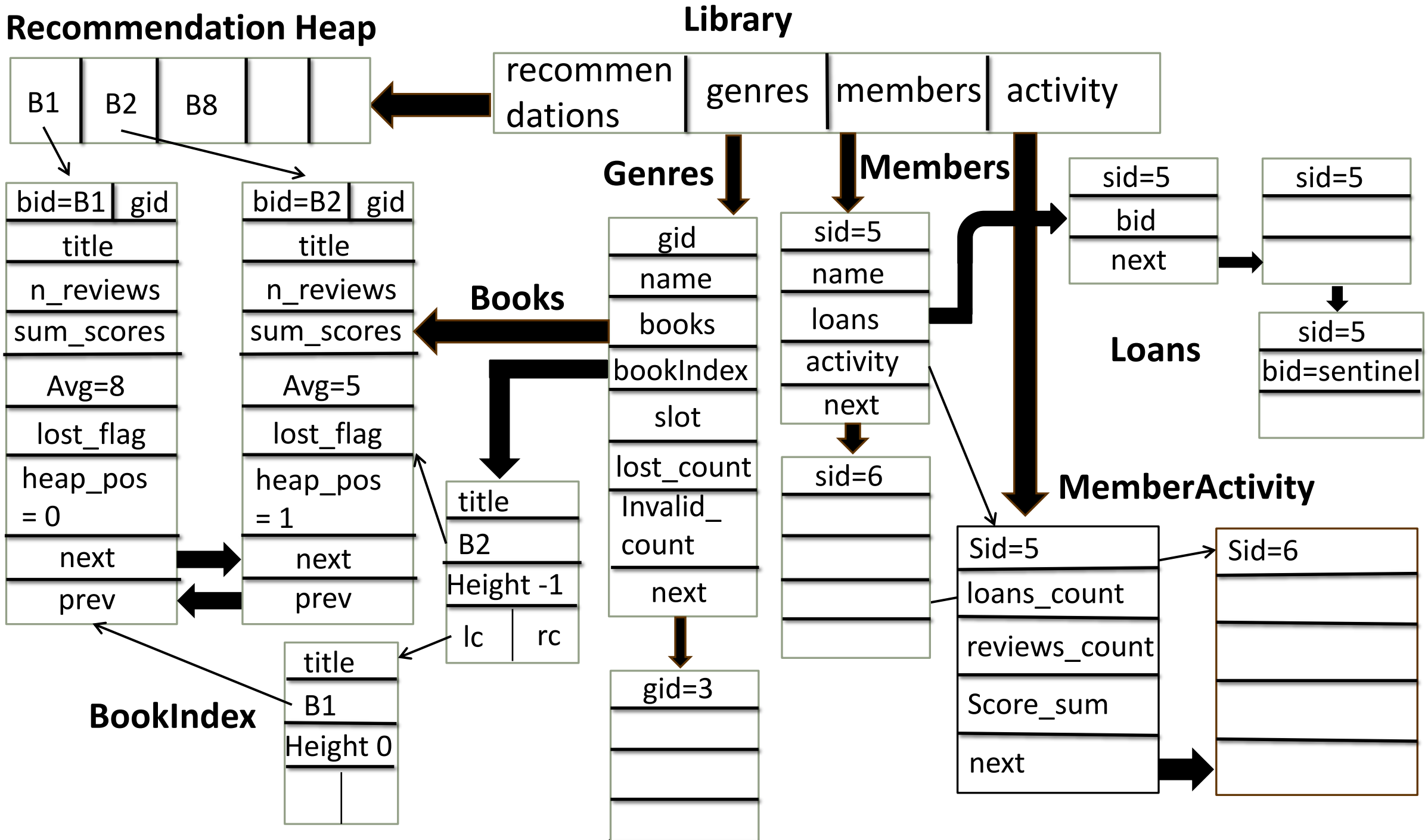
New Focus: Advanced search, recommendations, and statistical analysis

Enhanced Data Structures:

- Search Trees (AVL)
- Heaps (MaxHeap)
- Dynamic Activity Lists

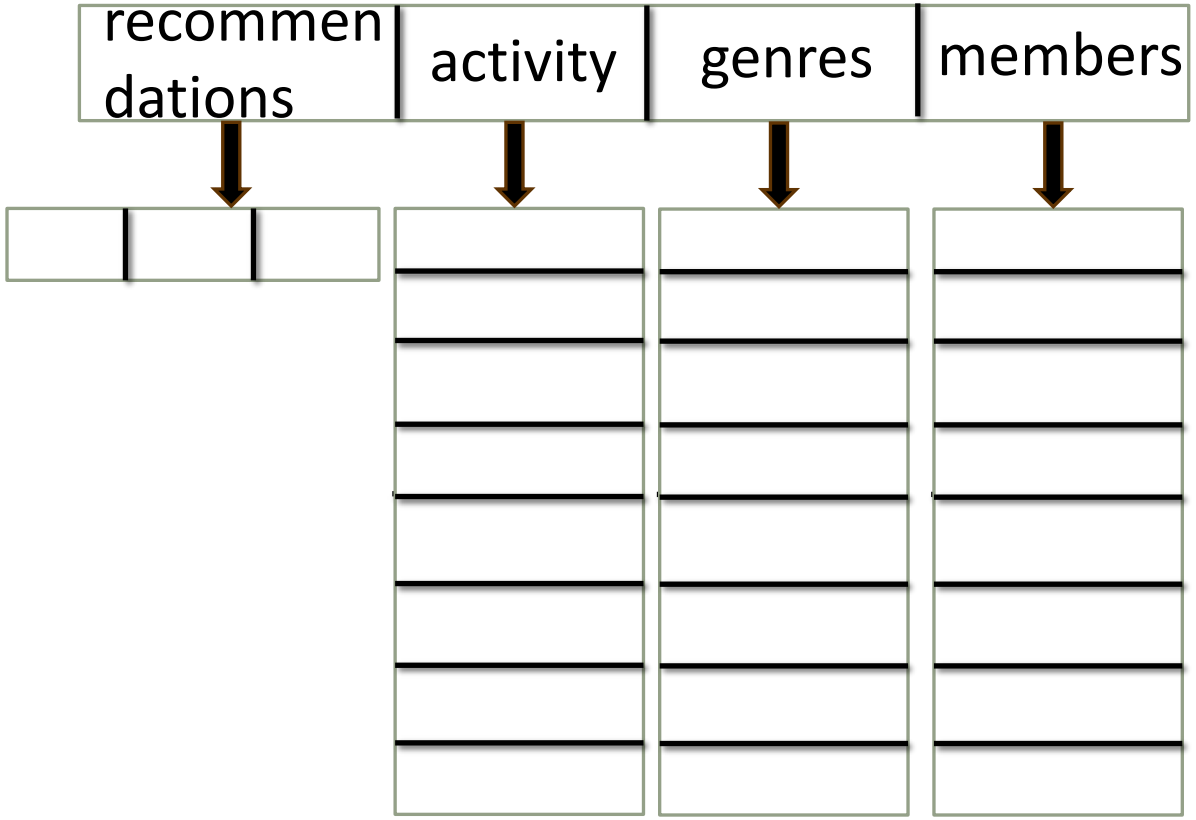
Goal: Efficient O(log n) operations for search and recommendations

Design & Data Structures


Recommendation Heap
B1
B2
B8
Library
recommen dations
genres
members
activity
Genres
Members
bid=B1
gid
title
n_reviews
sum_scores
Avg=8
lost_flag
heap_pos = 0
next
prev
bid=B2
gid
title
n_reviews
sum_scores
Avg=5
lost_flag
heap_pos = 1
next
prev
Books
gid
name
books
bookIndex
slot
lost_count
Invalid_ count
next
sid=5
name
loans
activity
next
sid=6
sid=5
bid
next
sid=5
sid=5
bid=sentinel
Loans
title
B2
Height -1
lc
rc
title
B1
Height 0
BookIndex
gid=3
MemberActivity
Sid=5
loans_count
reviews_count
Score_sum
next
Sid=6

Library

Library



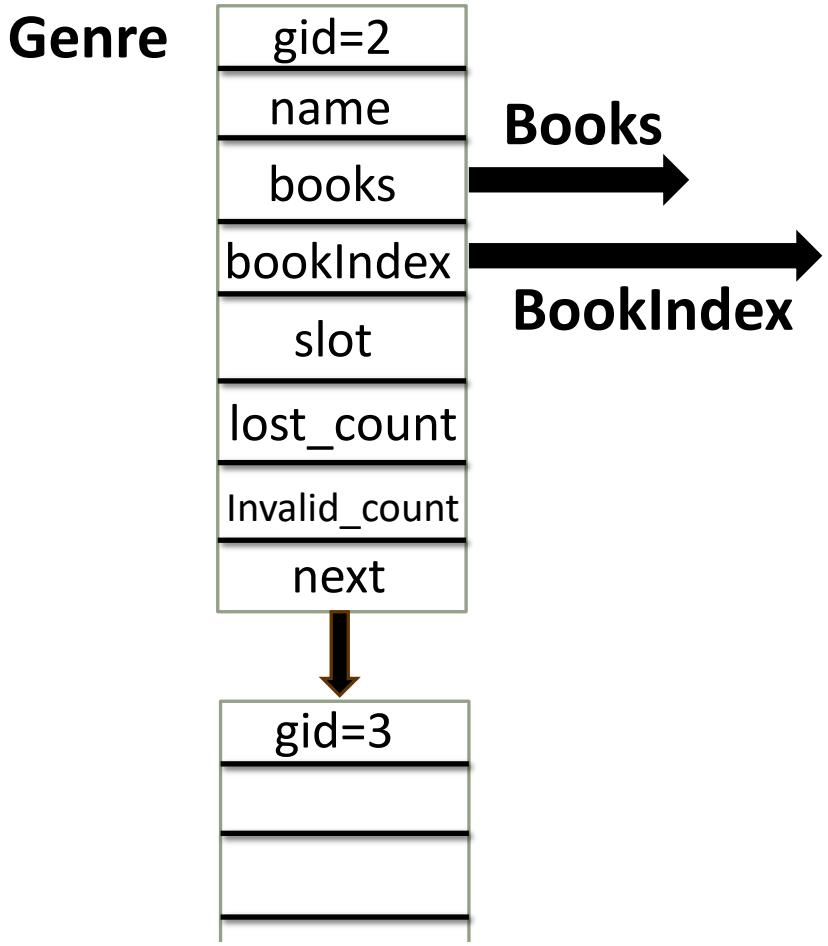
Type Library

- struct of 2 pointers to lists
- genres pointer to Genre List
- members pointer to Member List
- **activity** pointer to MemberActivity List
- **recommendations** pointer to Recommendation Heap

MemberActivity is not sorted

1 global Library instance

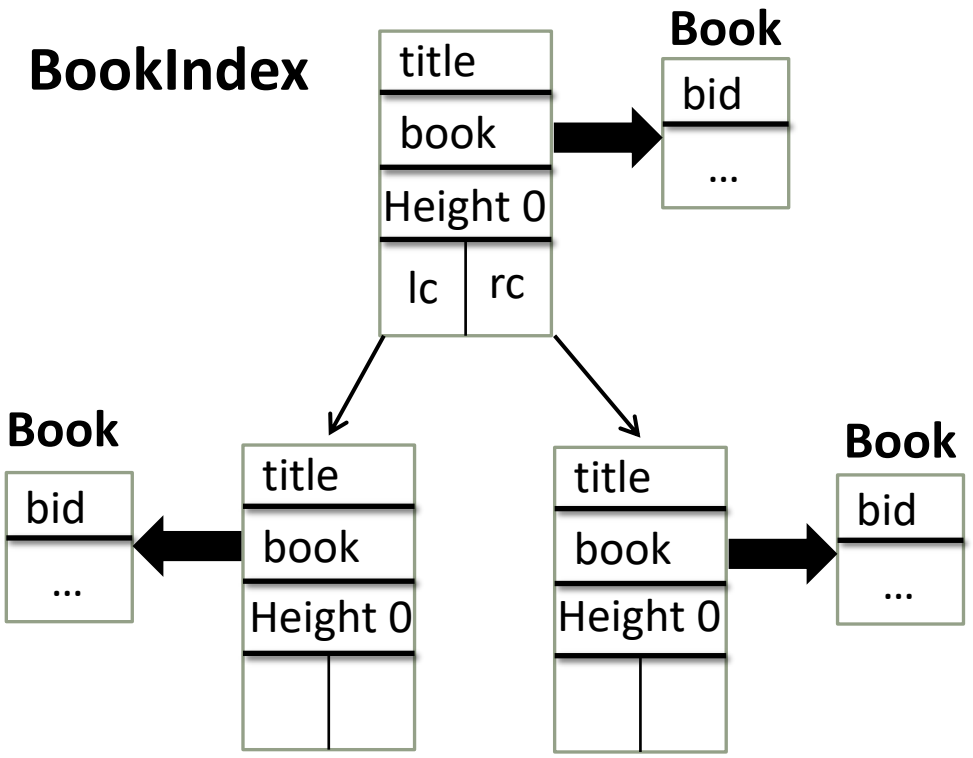
Genre List



Type Genre

- gid unique id
- name string (chars array)
- books pointer to Book List
- slots count of display slots
- lost_count count of lost books
- invalid_count count of invalid reviews
- next pointer to Genre
- **bookIndex** pointer to BookIndex

BookIndex AVL



BookIndex AVL

- AVL tree
- Lexicographically sorted by title

Properties:

- Unique titles only
- Updated on book insertion/deletion

Members:

- **Title** of the book
- **Book** pointer to the book
- **lc, rc** left, right child.
- **height** AVL height

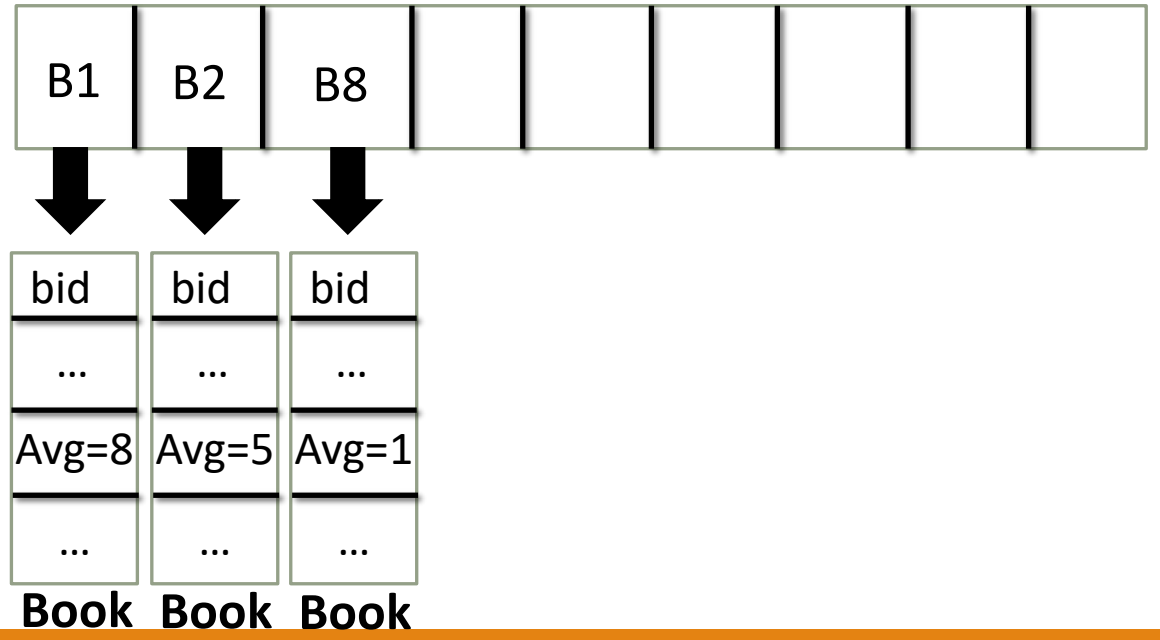
Complexity: O(log n) search/insert

Recommendation Heap

Recommendation Heap

Capacity: 9

Size:3



Recommendation Heap

- MAX Heap
- Sorted by **book avg**

Stores pointers to the top books regarding the genre

Updated When:

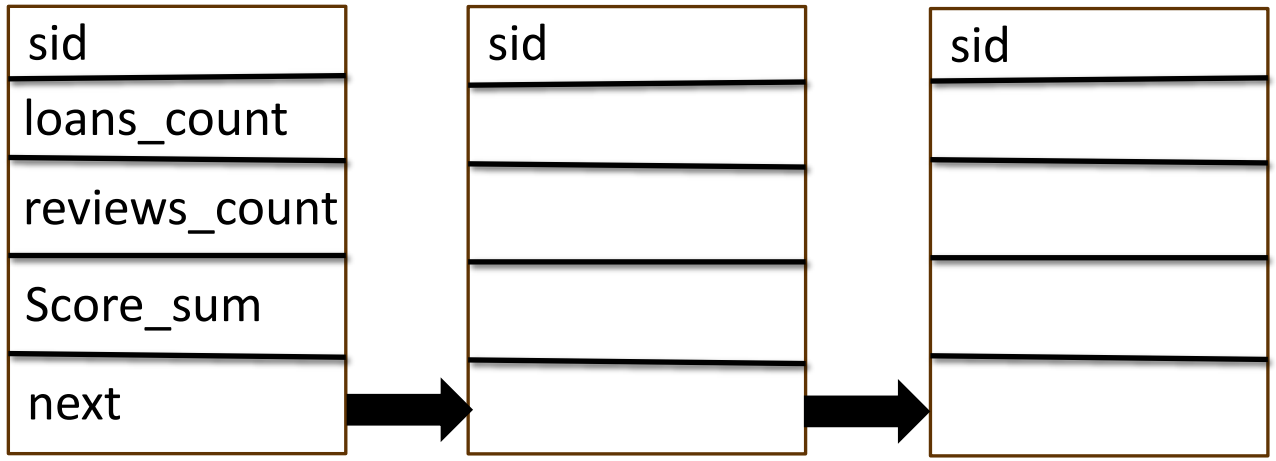
- Book return with rating change
- Book marked as lost (removed)

Type Member

- **heap** array
- **size** number of existing books
- **capacity** maximum size

MemberActivity List

MemberActivity



MemberActivity List

- Singly-Linked
- Unsorted

Tracks: Loans, reviews, and scoring activity

Updated on every **L** (Loan) and **R** (Return) operation

Complexity: O(1) inserts & updates

Type Member

- **sid** of the member
- **loans_count** #loans from sid
- **reviews_count** #reviews from sid
- **score_sum** Σ(score) from any given review
- **next** pointer to the next block

System Stats

Global variables or different struct or structs
extra members

Type

- **book_count** #books in the library
- **member_count** #members
- **active_loans** #active loans
- **all_score_avg** average number of all scores

Events

New Events

Command	Description	Output
1. F <title>	Search book by title using **BookIndex**	Print book info or NOT FOUND
2. TOP <k>	Show top k books from the **Max heap**	Print top books
3. AM	Show most active members using **MemberActivity**	Print Member stats or NO ACTIVE MEMBERS

New Events

Command	Description	Output
4. U <bid> "<new_title>"	Update book title using **BookIndex**	Print DONE/IGNORED
5. X	Print System statistics using **System Stats**	Print library stats
6. BF	Free all structures	DONE

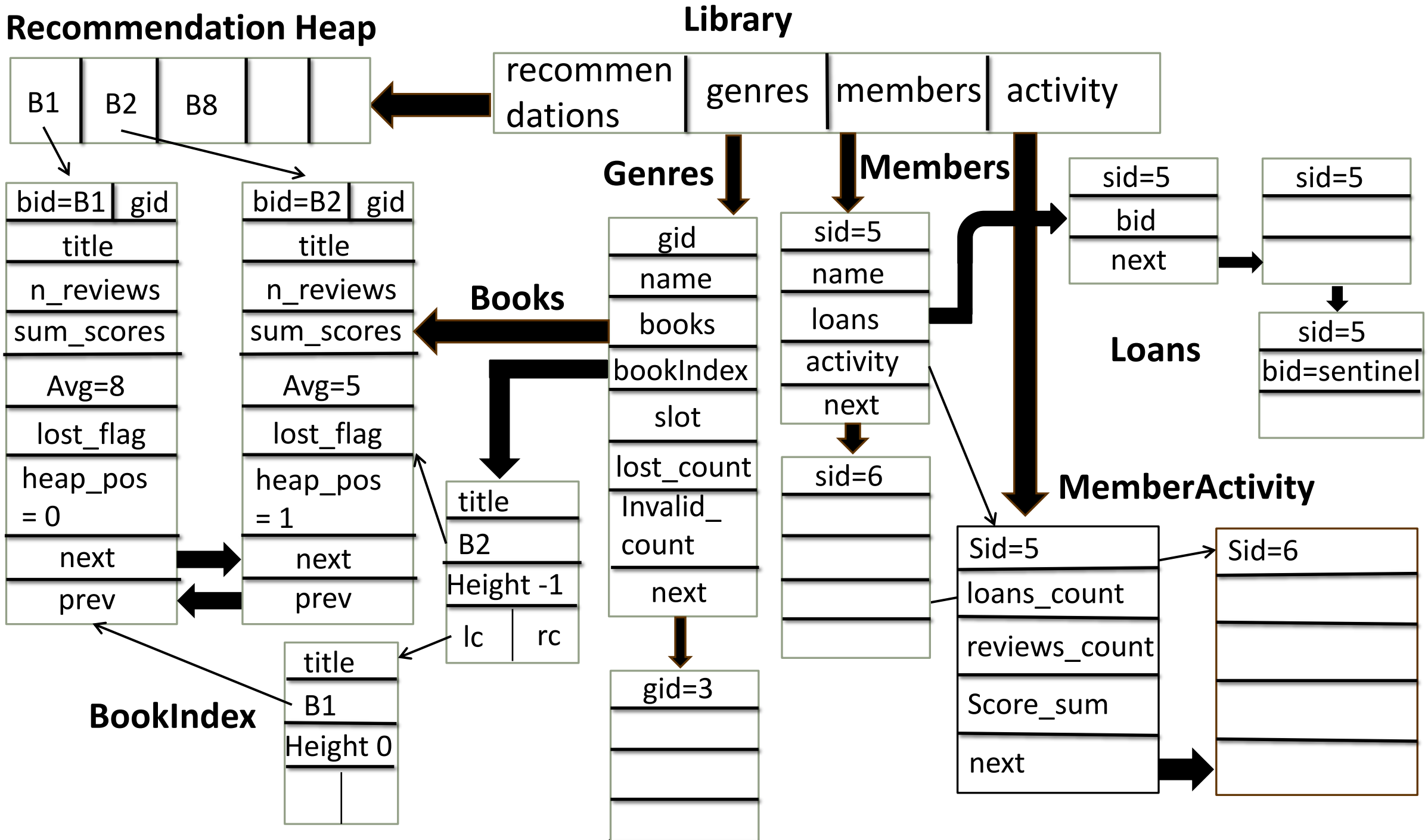
Examination Tips

Practice

- Explaining your algorithms using high-level non-technical language
- Try it on friends and non-technical people
  - Rubber duck method
- Make sure your descriptions are understandable and convincing

Thoroughly **review** your project the day before the examination

- Make sure your **remember** your design decisions and implementation details.
- Try to answer **why** you made a design decision


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B8
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recommen dations
genres
members
activity
Genres
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title
n_reviews
sum_scores
Avg=8
lost_flag
heap_pos = 0
next
prev
bid=B2
gid
title
n_reviews
sum_scores
Avg=5
lost_flag
heap_pos = 1
next
prev
Books
gid
name
books
bookIndex
slot
lost_count
Invalid_ count
next
sid=5
name
loans
activity
next
sid=6
sid=5
bid
next
sid=5
sid=5
bid=sentinel
Loans
title
B2
Height -1
lc
rc
title
B1
Height 0
BookIndex
gid=3
MemberActivity
Sid=5
loans_count
reviews_count
Score_sum
next
Sid=6